

Eyesight

Animals use senses to avoid danger, to find food, and also to find breeding partners. They may rely on touch, taste, and smell, depending on what they are investigating. For many predators, eyesight, or vision, is the most important sense. Eyes can be smaller than a period or more than 20 in (50 cm) across. Some animals see the world in shades of gray, but many can detect colors, including some that are invisible to us. All this information is passed along nerves, which connect the eyes to the brain. Here, the signals are processed, so that an animal can make sense of what it sees.

Look before you leap

This jumping spider pounces on small insects, spotting them with four pairs of eyes. It uses two pairs of sideways-pointing eyes to detect any movement that could spell food. If it sees something move, it turns toward it and focuses with its extra-large median eyes, which give it a very detailed image. Finally, the spider uses two smaller lateral eyes to judge the distance, before suddenly leaping on its prey.

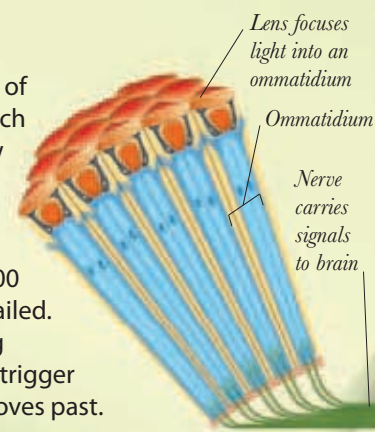


All-around view

Dragonflies hunt by sight, chasing other insects through the air. With their flexible necks and large bulging eyes, they can see in almost any direction when they fly. Like all arthropods, they have compound eyes, which are built in a totally different way from the eyes of mammals and other vertebrates. Dragonflies also have three much smaller, simple eyes, or ocelli, which sense the overall level of light.

Compound eyes

The eyes of arthropods are made up of compartments called ommatidia. Each ommatidium senses part of the view and forms a separate image. The animal's brain puts these images together, creating a picture like a mosaic. Dragonflies have up to 30,000 ommatidia, so their pictures are detailed. Compound eyes are good at sensing movement, because the ommatidia trigger in quick succession as something moves past.



Median eye can swivel to keep the prey directly in the line of sight

Sideways-pointing eyes usually spot prey first

Lateral eye is forward facing and set far apart from the other one, giving good distance judgment